

How much energy does an AI operation need?

core:

The core compute energy of the AI operation is approximately constant, but the total environmental footprint varies by serving data center, infrastructure efficiency, electricity mix, cooling water intensity, and network path.

Compute energy path:

1. Model computation (largest share)

This is the energy used by GPUs/TPUs running the model.

For example:

generate image → diffusion model inference → GPUs compute

This part is **almost identical everywhere** because the model performs the same operations.

Typical share:

~80–95% of total energy

2. Data center infrastructure overhead

Data centers require additional energy for:

- cooling
- power conversion
- networking
- storage

This overhead is expressed with **PUE (Power Usage Effectiveness)**.

Typical values:

Data center type	PUE
Efficient hyperscale	~1.1
Average facility	~1.3–1.5

Energy consumed becomes:

Total energy = compute energy × PUE

So the same AI task could consume slightly different amounts depending on the data center efficiency.

3. Network transmission

Energy is also used to move data between:

user → servers → data center

But this is usually **very small compared to computation**.

Typical share:

<5% of total energy

Total electricity:

Total serving electricity is the total electricity used by the infrastructure to execute and deliver an AI request.

<https://cloud.google.com/blog/products/infrastructure/measuring-the-environmental-impact-of-ai-inference/>

Energy:

Total electricity (kWh) = compute + infrastructure (PUE) + network

- **total electricity = (compute × infrastructure multiplier) × network**

Impact:

Environmental impact = total electricity × local factors (carbon, water)

- **CO₂ = total electricity × carbon factor of serving region**
- **water = total electricity × water factor of serving region**

Compute energy per operation:

Operation	AI Type	Compute Characteristics	Core Compute Energy per Action (kWh)	Relative Intensity
Chat with AI (single prompt)	NLP / LLM inference	Short text input/output, moderate model size	0.0003 – 0.002 kWh	Low
Real-time translation (voice/camera, ~10–30 sec)	Speech + Vision + NLP	Continuous streaming inference, multi-model pipeline	0.001 – 0.01 kWh	Medium
Summarize PDF document	Long-context LLM inference	Large input tokens, memory-intensive processing	0.002 – 0.02 kWh	Medium–High
Remove background from image	Computer vision (segmentation)	Single-pass vision model, relatively efficient	0.0005 – 0.005 kWh	Low–Medium

This table shows how much raw computing energy each of the four AI operations requires. Chat with AI is the lightest, consuming roughly 0.0003 to 0.002 kWh per prompt, because it only runs a text model on a short input. Real-time translation uses more — around 0.001 to 0.01 kWh — because it combines speech, vision, and language models running continuously. Summarizing a PDF sits between 0.002 and 0.02 kWh, since the model has to process a large amount of text at once. Background removal is relatively efficient at 0.0005 to 0.005 kWh, because it uses a single focused computer vision model. The last column labels each operation as low, medium, medium-high, or low-medium in terms of relative energy intensity, giving a quick visual ranking.

Logic per operation:

Operation	Location Logic
Chat	hub → stable infrastructure and connectivity
Real-time translation	Closest region → minimize latency (continuous stream)
PDF summarization	Strong compute hub → maximize capacity (heavy inference)
Image processing	Fast API hub → optimize response speed

This table explains the reasoning behind which type of data center is best suited for each operation. Chat is sent to a balanced, stable hub because it needs consistent infrastructure. Translation is routed to the closest available region because even a small delay makes the live experience feel broken. PDF summarization goes to the most powerful compute hub available because processing a long document requires maximum capacity. Background removal is sent to the fastest API hub to minimise response time for the user. This table essentially explains the "why" behind the routing choices shown in the city-specific sections.